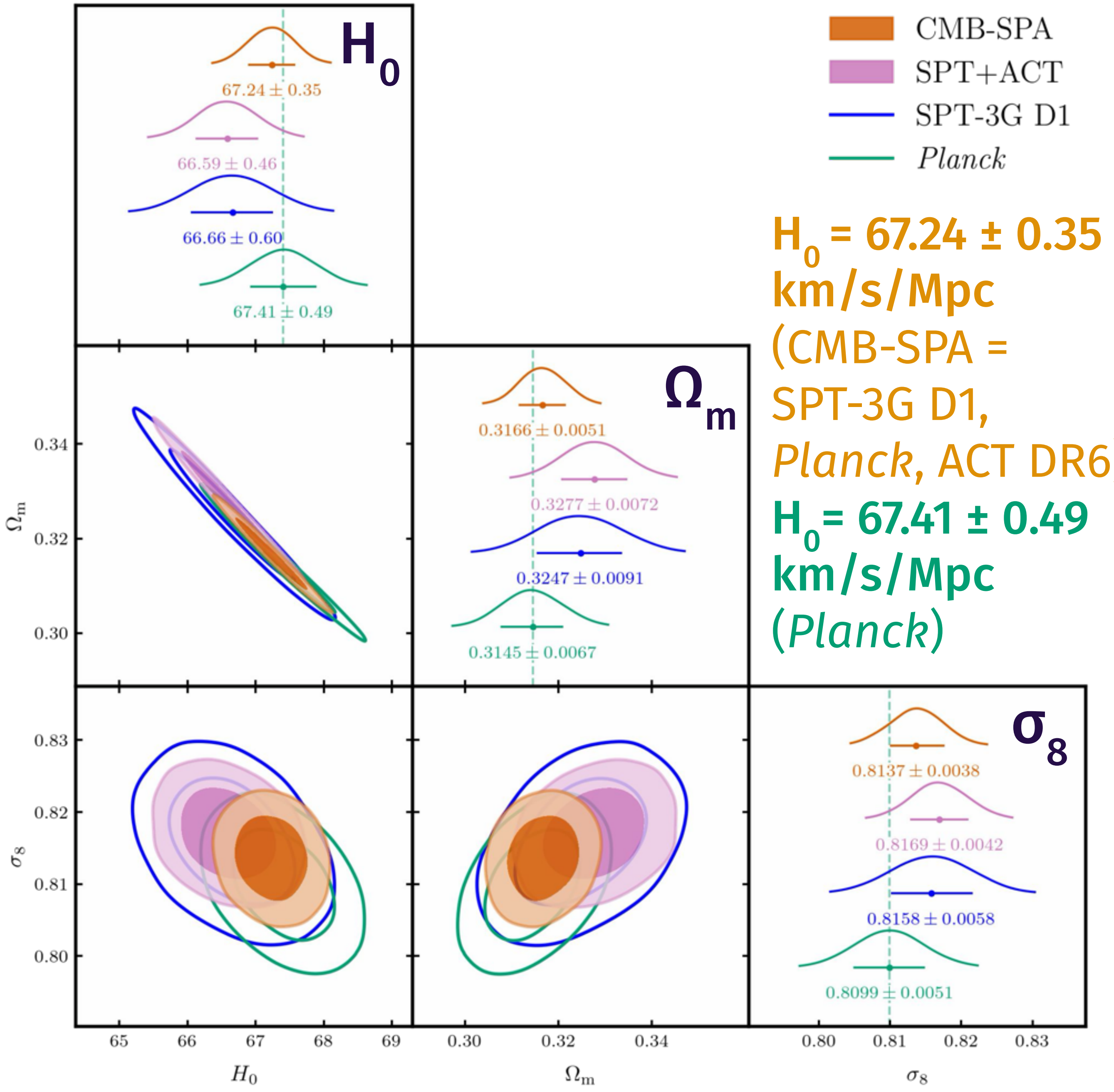
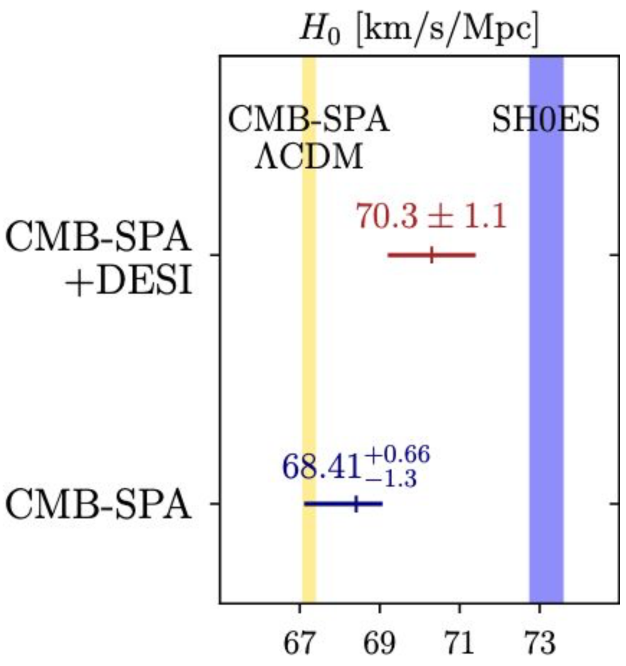


Most precise constraints on ΛCDM from CMB to date

H_0 = 67.24 ± 0.35 km/s/Mpc (CMB-SPA = SPT-3G D1, *Planck*, ACT DR6)

H_0= 67.41 ± 0.49 km/s/Mpc (*Planck*)

- All three experiments (SPT-3G D1, *Planck*, ACT DR6) agree within 1.1σ



- Hubble tension with SH0ES persists
- Model expansion including axion early dark energy does not resolve; DESI combo helps

NB: 2.3 sigma tension with DESI DR2 in the context of LCDM

Slide from Ali Rida Khalife, Right: Fig. 3 Khalife+ arXiv:2507.23355

ACT+SPT+Planck combined CMB lensing

"Unified and consistent structure growth measurements from joint SPT, Planck and ACT CMB lensing"

Combines lensing from the two ground experiments to put the tightest constraints on the growth of structure and sound-horizon-independent estimate of H_o

$\sigma_8 = 0.829 \pm 0.009$

$H_0 = 66.4 \pm 2.5 \mathrm{kms}^{-1}\mathrm{Mpc}^{-1}$

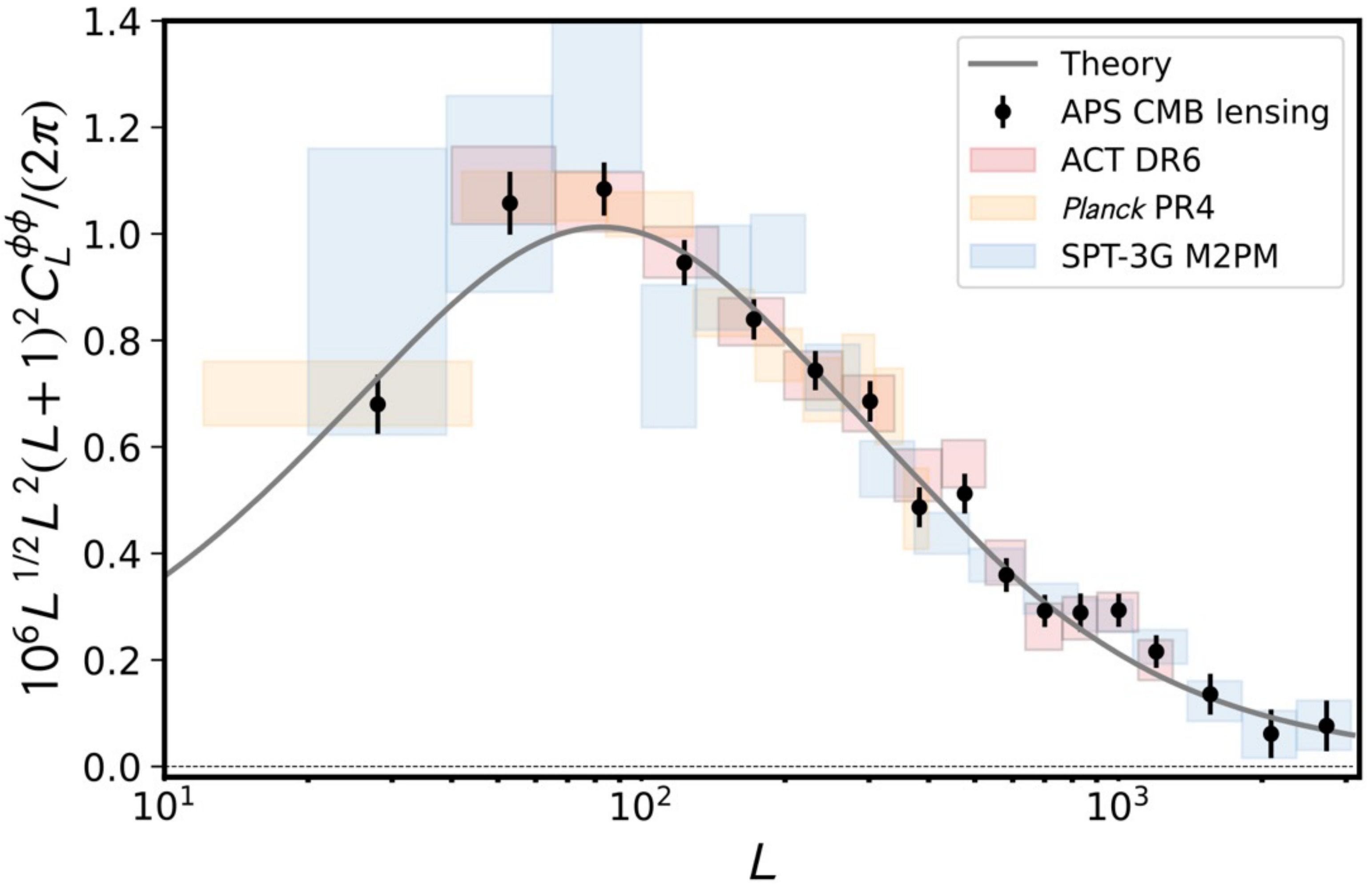
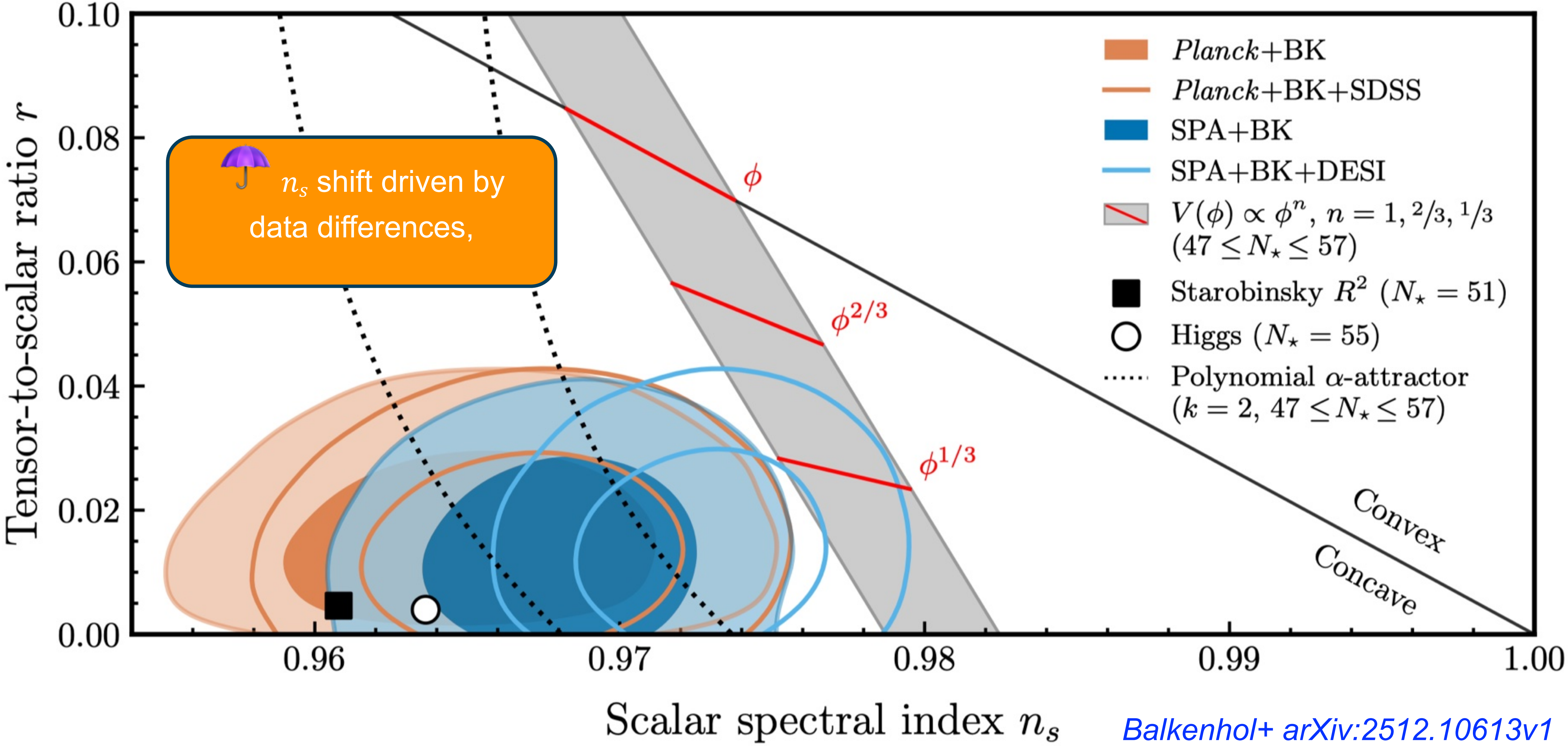


FIG. 2. We present the combined lensing bandpowers from the three surveys in black. In the background we show the *Planck* lensing bandpowers from PR4 NPIPE analysis in orange, the ACT DR6 lensing potential power spectrum bandpowers in red, and the lensing bandpowers from SPT-3G M2PM in blue. The gray line shows the theory prediction from the best-fit cosmology of the CMB likelihood. Note that we have applied an additional $L^{1/2}$ scaling over that usually used to display bandpowers to enhance visually the small scales.

Qu, Ge, et al., arXiv:2504.20038

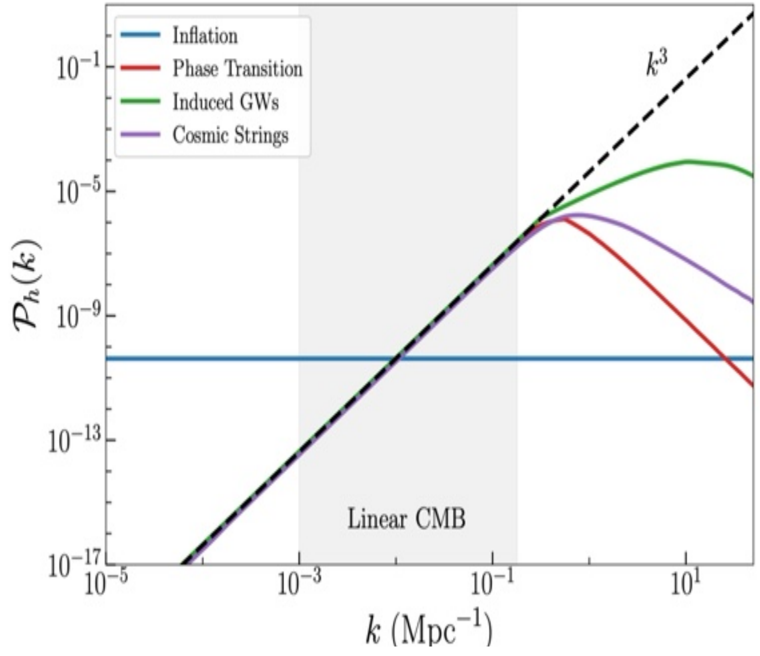
Inflation at the End of 2025: Constraints on r and n_s Using the Latest CMB and BAO Data

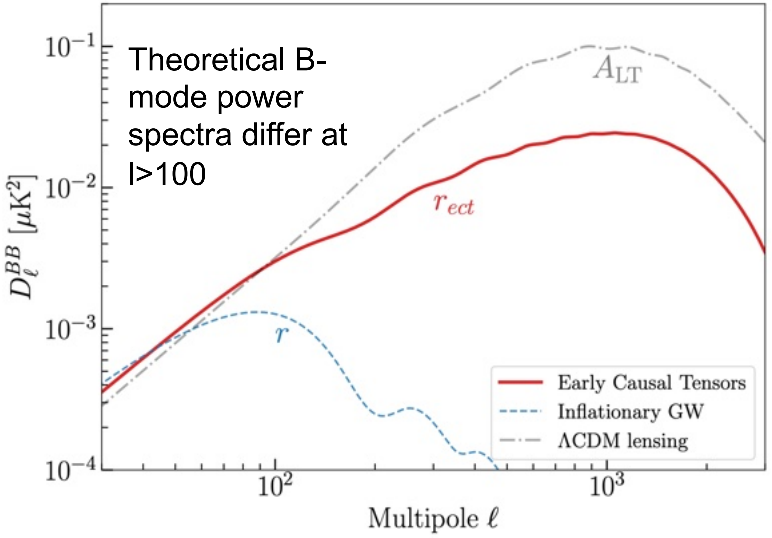


Balkenhol+ arXiv:2512.10613v1

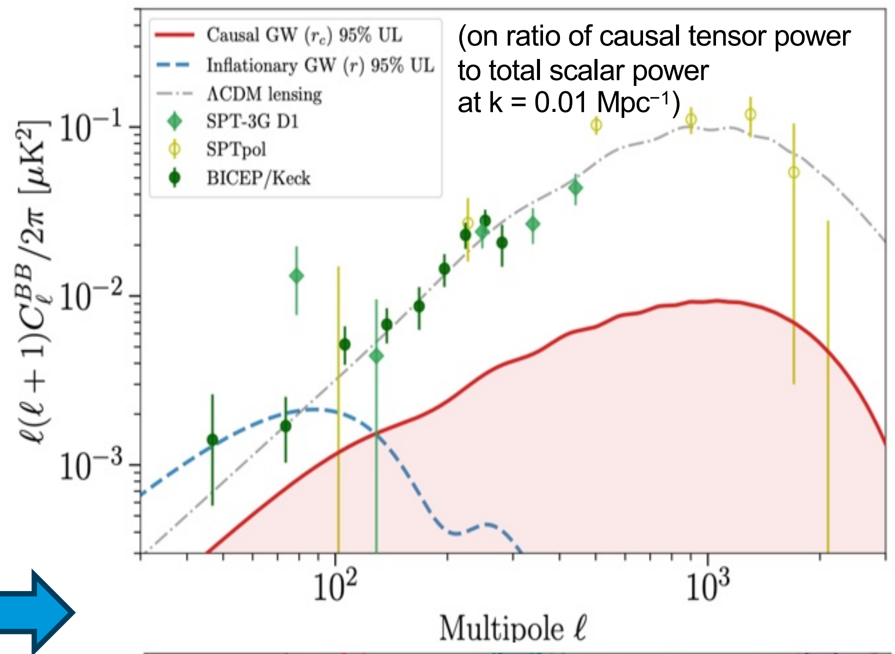
Early Causal Tensor (ECT) sources of fluctuations

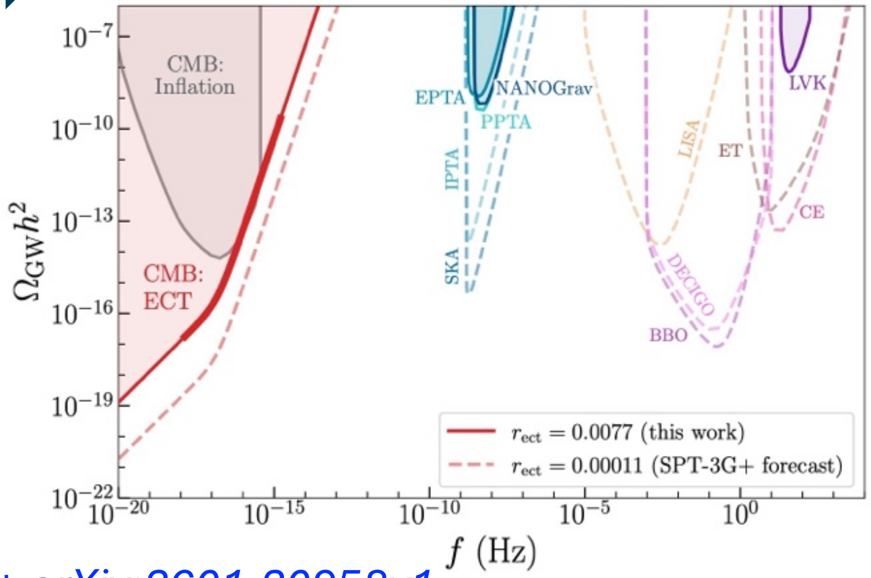
1. Post-inflationary sources are fundamentally limited by the impossibility of "super-horizon" correlations
2. Thus, they universally yield white-noise spectra on sufficiently large scales. For all such sources, the dimensionless tensor power spectrum exhibits a characteristic $P_h(k) \propto k^3$ scaling at small comoving wavenumber k.
3. This robust, model independent consequence of causality and finite correlation lengths holds universally for length scales larger than the physical extent of the source, provided that the latter is bounded in frequency and time [Cai, Pi, Sasaki, PRD 102, 083528 (2020), arXiv:1909.13728].
4. If the tensor source is before $z \sim 7 \times 10^4$, this turn over will be hidden in the CMB by Silk damping (the horizon size at $z \sim 7 \times 10^4$ matches the Silk damping scale).





SPT+BK ➔ r_{ect} < 0.0077 95% CL upper limit



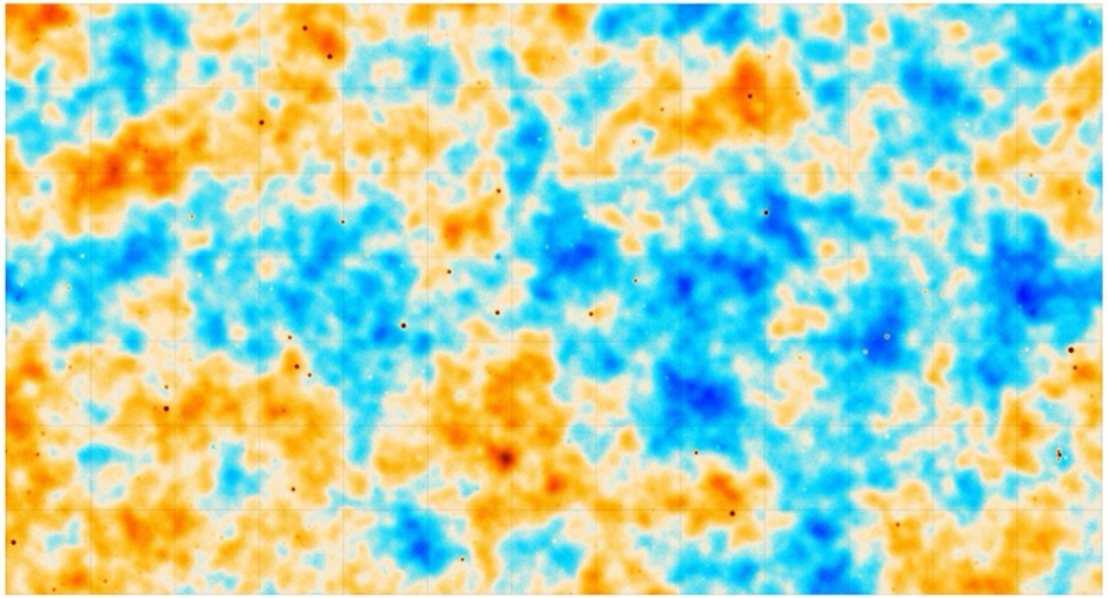


Zebrowski+ arXiv:2601.20958v1

A promising near-future for CMB science

The next years will see the emergence of new CMB datasets both from the Atacama desert with **Simons Observatory** and future releases from the **South Pole Telescope**

SO currently collecting data !!



credit: SO collaboration

credit: POLOCALC/HoverCal drone team

(Slide La Posta at Pont d'Avignon)

Future SPT constraints

The full 10 000 deg^2 SPT-3G dataset is expected to achieve an improvement in the ΛCDM FoM of 130; i.e., SPT-3G will reduce the allowed six-dimensional parameter volume by a factor of 130 compared to Planck.

This reduction in volume is comparable to what is expected from the first two years of the SO-Baseline dataset (which comes next).

This similarity in constraining power, plus the substantial amount of overlap in sky coverage, will provide an opportunity for significant map-level consistency tests. These can be used to detect, or limit, sources of systematic error, increasing the robustness of the parameter determinations from both SO and SPT3G.

Prabhu et al. arXiv:2403.17925v3

See realistic expectations on parameters in Vitrier+ arXiv:2510.24669v2

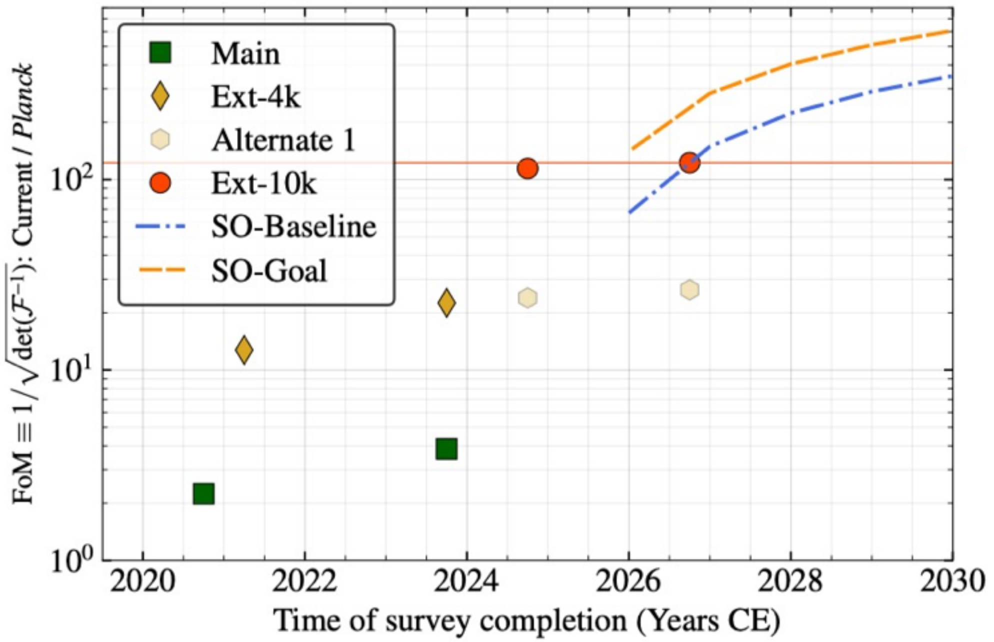
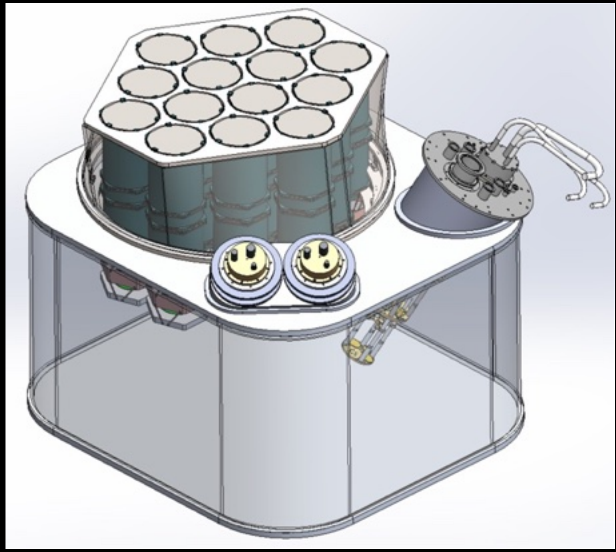


Figure 7. Figure of merit (FoM) for ΛCDM compared to *Planck*: We present the FoM for SPT-3G Main (green), Ext-4k (yellow), and Ext-10k (red). The transparent yellow circle represents the improvement in Ext-4k, if we were to continue the SPT-3G Main survey during 2024 as opposed to conducting the SPT-3G Wide survey. The evident increase in FoM with the new strategy emphasizes its potential efficacy over continuing the SPT-3G Main survey for cosmological parameter constraints. For reference, we also show the FoM expected for the two SO configurations (Baseline in blue and Goal in orange) under the assumption that the observations will begin in early 2025.

Upcoming new SPT-3G+ camera

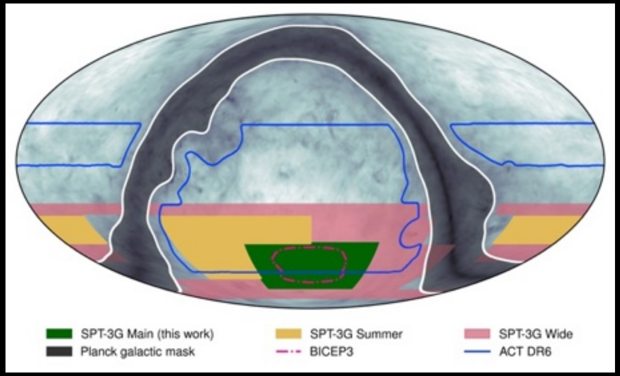
SPT-3G+ camera planned to deploy in 2029
- Optimized for delensing
  - dichroic (90+150 GHz) dual-pol detectors at 100 mK
  - improved optics, pixel packing
- Greater than 7 times higher mapping speed than SPT-3G
- Projected 6 year map depths (in total intensity) over 800 deg^2:
  - 0.70 uK-arcm at 90 GHz
  - 0.68 uK-arcm at 150 GHz

    → combined 0.49 uK-arcm !



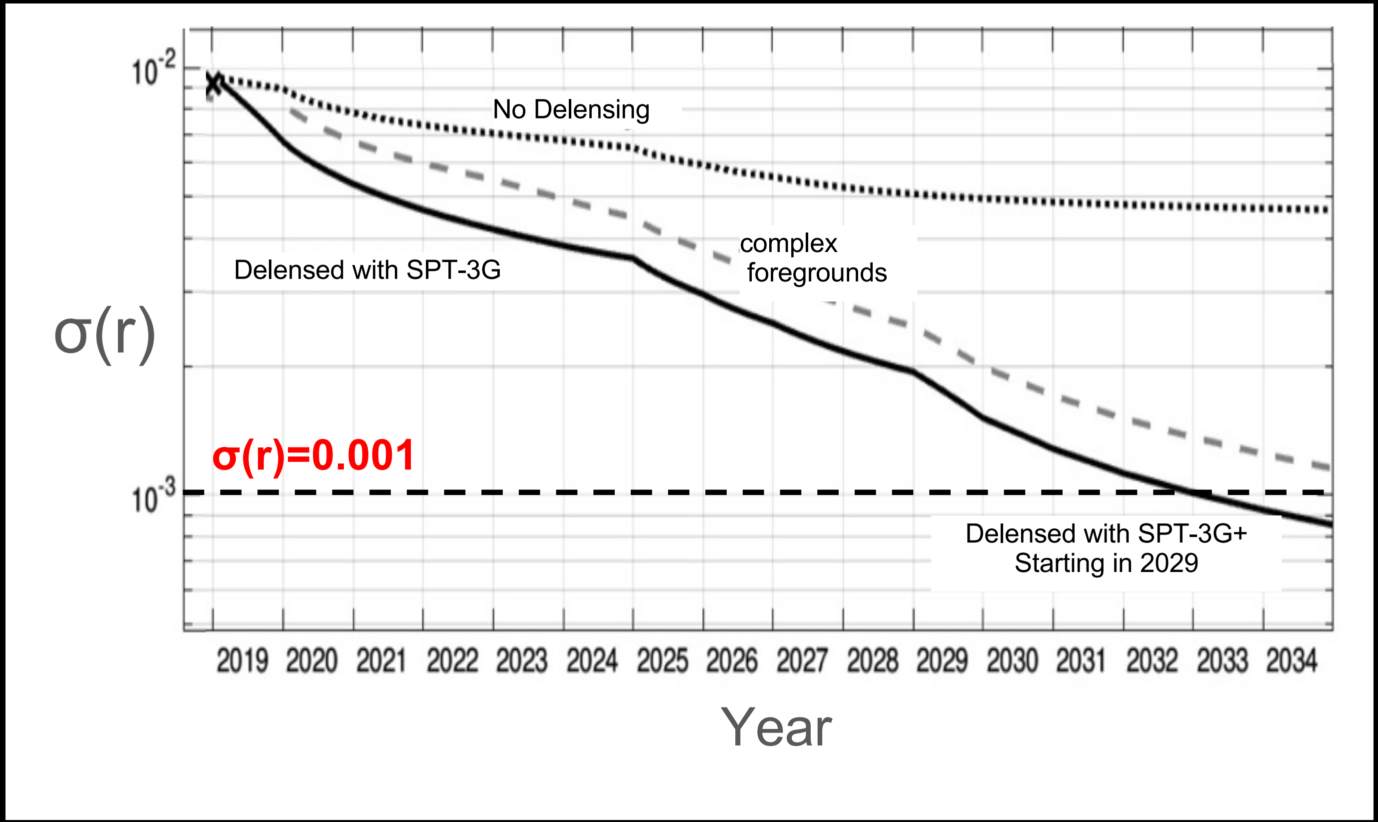
SPT-3G+ delensing with BICEP (SPO) to achieve **σ(r) ~ 0.001**
- Could be even lower with SPT-3G+ contributing to low-ell BB (e.g., Zebrowski arXiv: 2505.02827)

Also advances CMB / SZ science more generally
- CMB Cluster lensing: S/N ~ 1 for $10^{14} M_\odot$;
  ~ 10 for $10^{15} M_\odot$
- Could perform a 1-year 10,000 deg^2 "wide" survey to map depth of 4 uK-arcm, extending deep Southern CMB coverage.

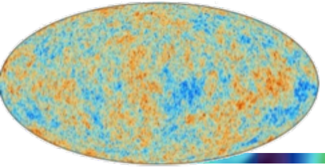


SPO w/SPT-3G+

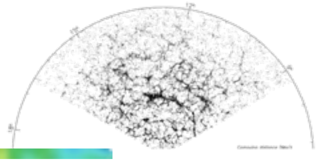
σ(r) reach of SPO with SPT-3G+



These forecasts are anchored on the demonstrated performance of BICEP and SPT measurements over the last decade



Other ongoing experiments…



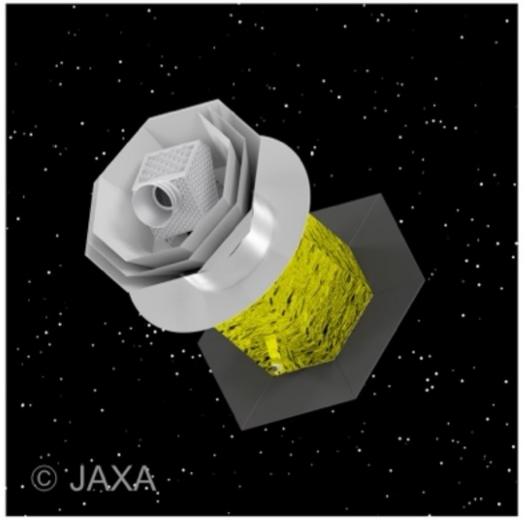
- On the ground
  - In the Canary Island (Quijote, LSPE/STRIPE [it], GroundBird)
  - At Llama site in Argentina: QUBIC (bolometric interferometer, IN2P3-led)
  - In Tibet: ALI (1 then 2)
  - Maybe at Hanle in Northern India ?
  - and/or Greenland ?, also in
  the northern hemisphere
- Balloons…
  - EBEX, SPIDER, PILOT… (like space, on the cheap, *when it works*!)

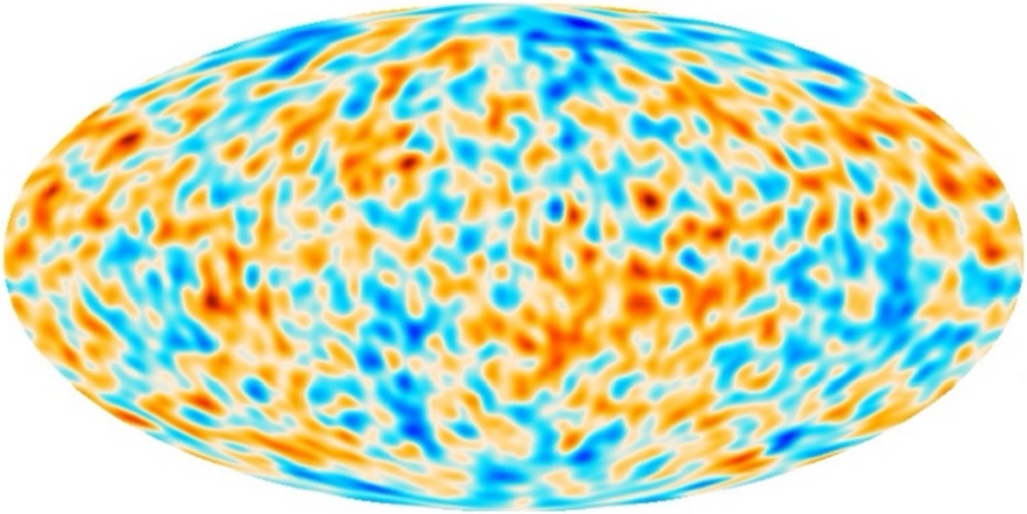
LITEBIRD

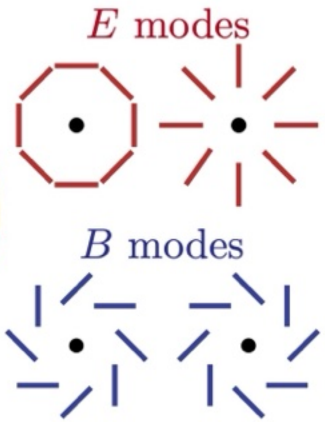
LiteBIRD overview

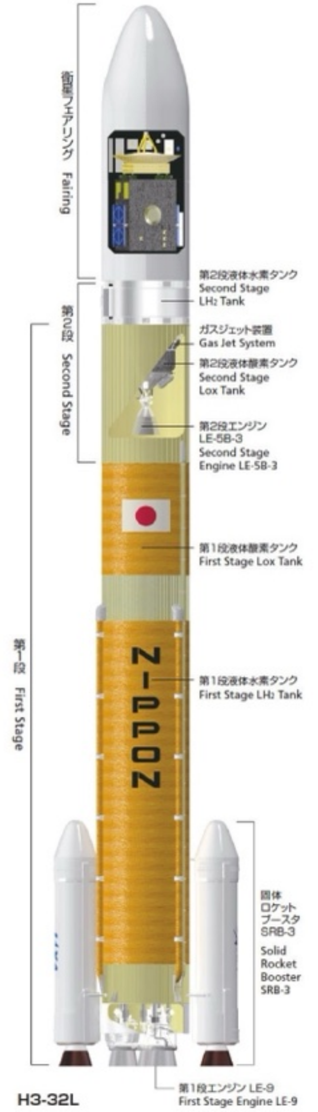
- Lite (Light) spacecraft for the study of *B*-mode polarization and Inflation from cosmic background Radiation Detection
- JAXA's L-class mission was selected in May 2019 to be launched by JAXA's H3 rocket on JFY2036
- Scientific goals: create all-sky, multi-band ultra-sensitive **microwave polarization maps**, test **inflationary models** through the measurements of **primordial gravitational waves**, and provide new insights into cosmology, particle physics and astrophysics
- **All-sky 3-year survey**, from Sun-Earth Lagrangian point L2
- Large frequency coverage (**40–402 GHz**) at **70–18 arcmin** angular resolution for precision measurements of the **CMB *B*-modes**
- Final combined sensitivity: **2.2 μK·arcmin**

LiteBIRD collaboration PTEP 2023









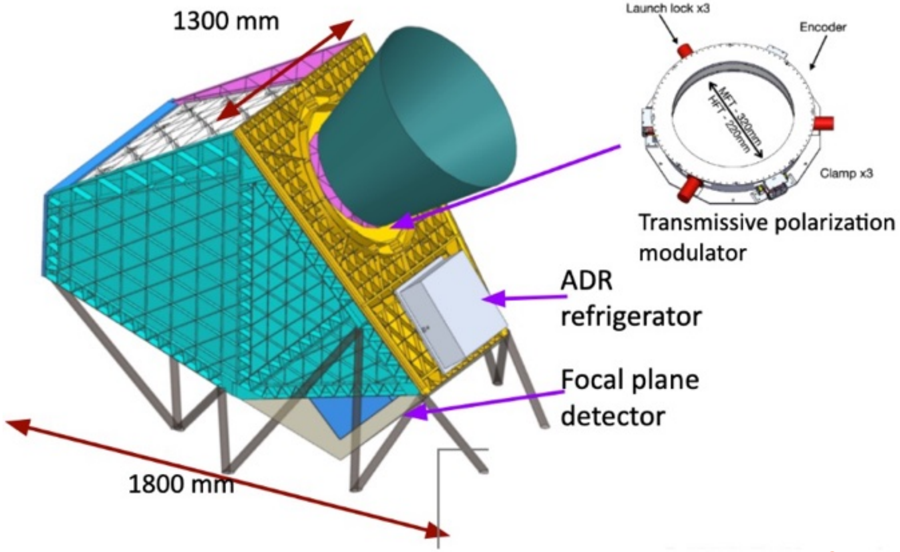
LiteBIRD reformation phase

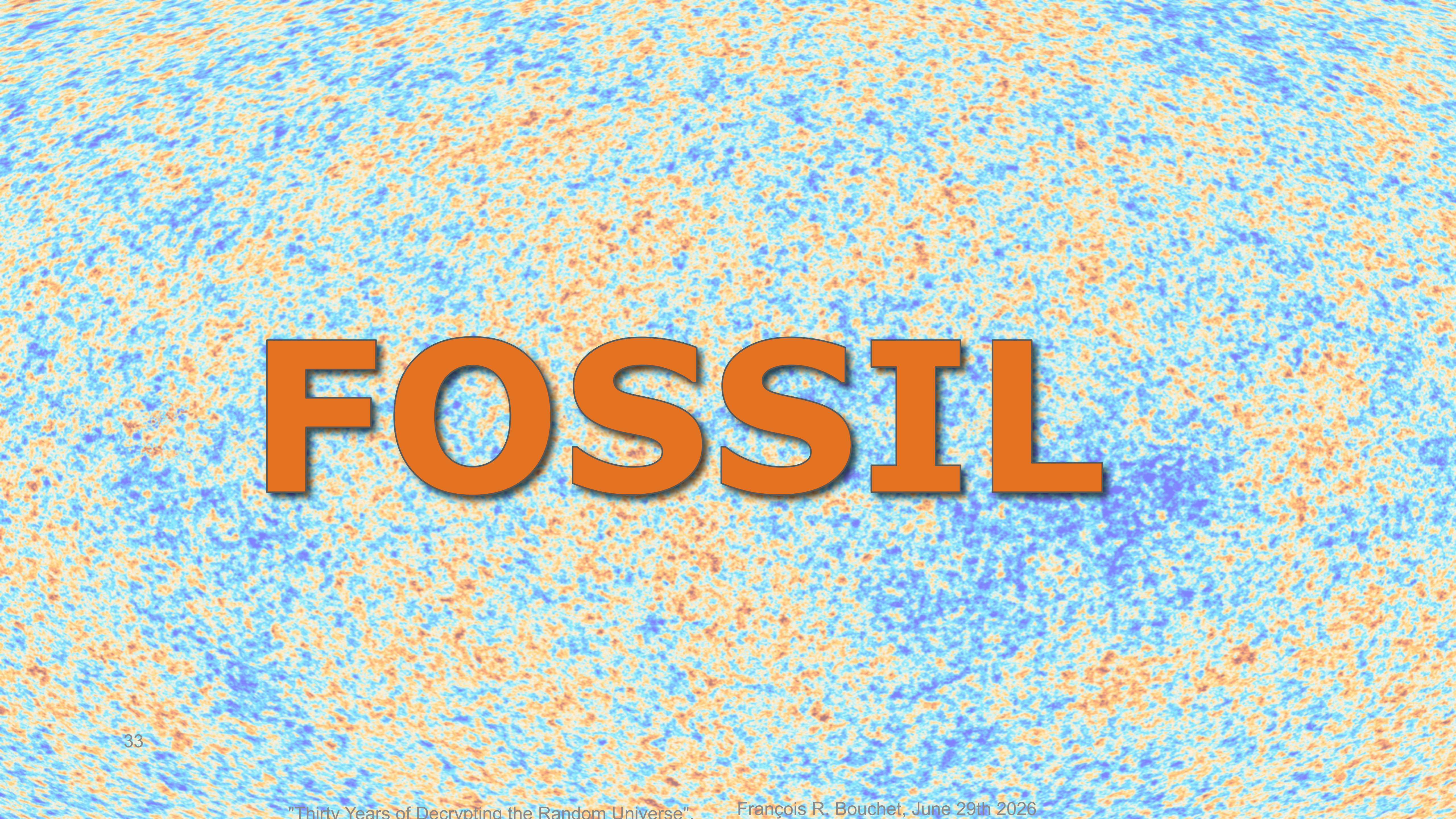
- *LiteBIRD* has been under **rescope studies** **to consolidate the mission's feasibility while keeping the same scientific objectives**
  - Revisit the error budget
  - **Simplify the mission configuration** (one single telescope instead of three; try to use existing technologies)
  - Simplify the cryogenic chain
  - Detectors to be procured by Europe
  - New HWP design based on stacking 6 plates in Pancharatnam configuration, providing large bandwidth
- JAXA Key Decision Point #2 passed successfully (Sept 2025)
- JAXA Mission Definition Review #2 scheduled for July 2026
- Targeted launch date JFY2036

  - **Two different configuration options** now being considered, both based on single Crossed-Dragone reflective telescope
  - Option 1 (no HWP) requires a faster spin rate to minimize 1/f noise
  - Option 2 is based on the possibility of using a wider-band HWP

Option 1
- Aperture 500 mm
- 40-570 GHz
- No HWP
- Spin rate 0.3 rpm

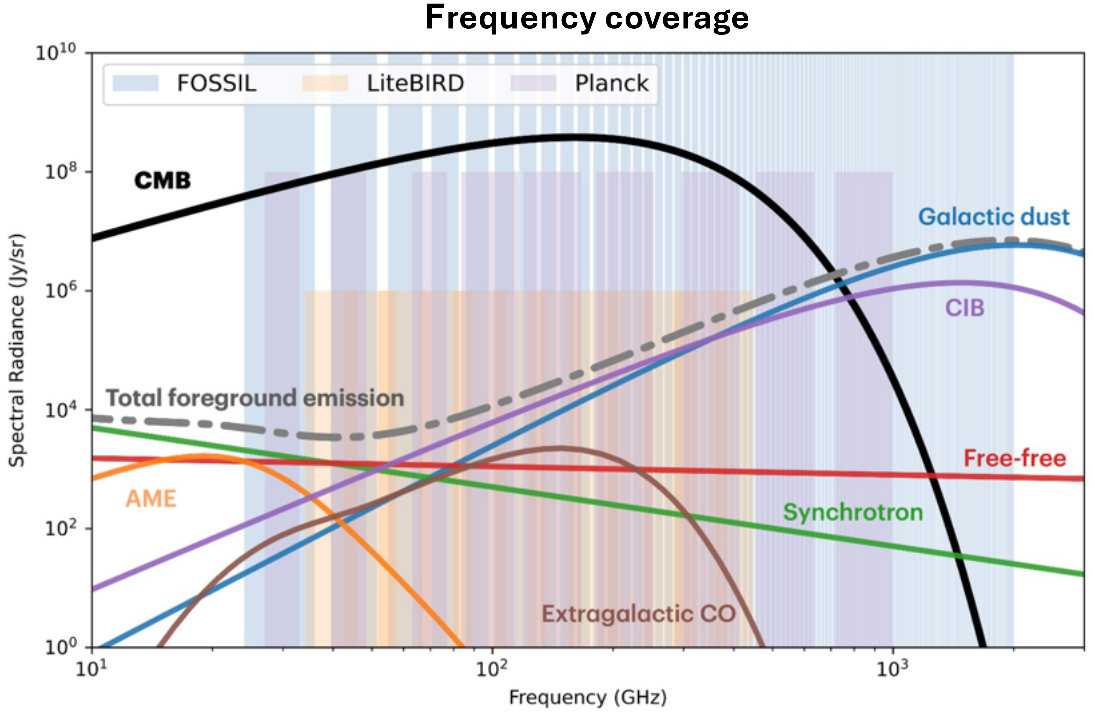
Option 2
- Aperture 500 mm
- 40-402 GHz
- Transmissive HWP
- Spin rate 0.05 rpm



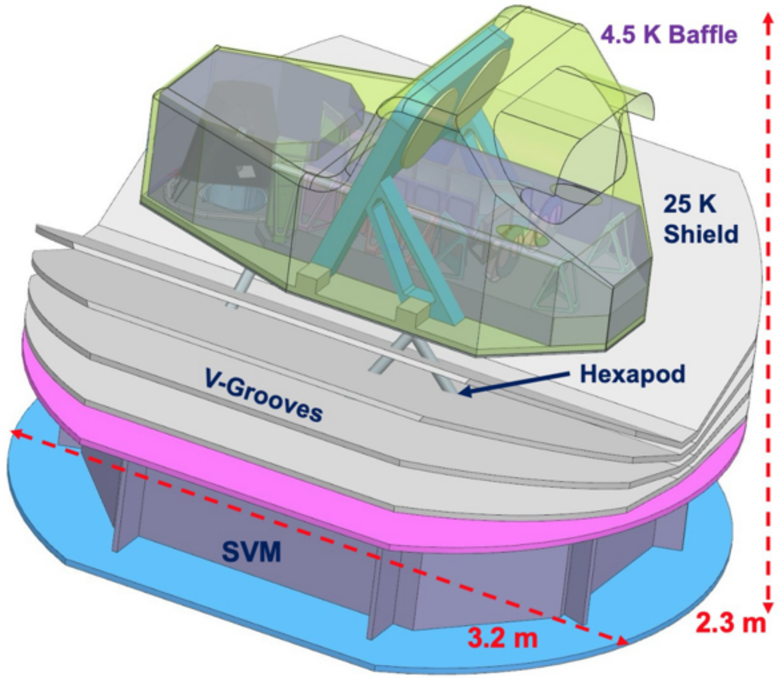

FOSSIL

FOSSIL IN RESPONSE TO ESA-M8 CALL (PI NABILA AGHANIM)

Credits: FOSSIL proposal 2026



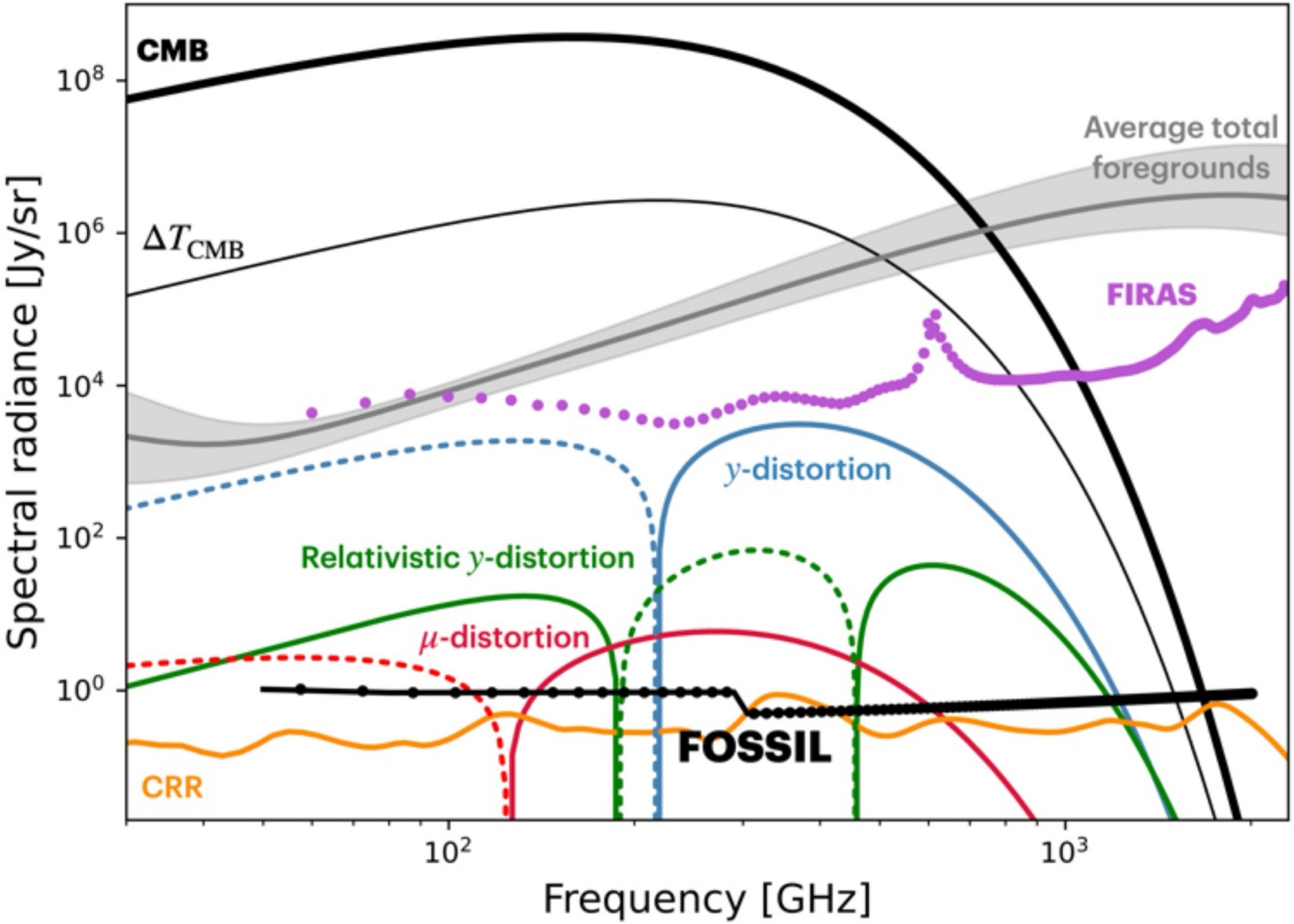
Credits: FOSSIL proposal 2026



50 years after FIRAS, a **full-sky absolute spectrometric survey**

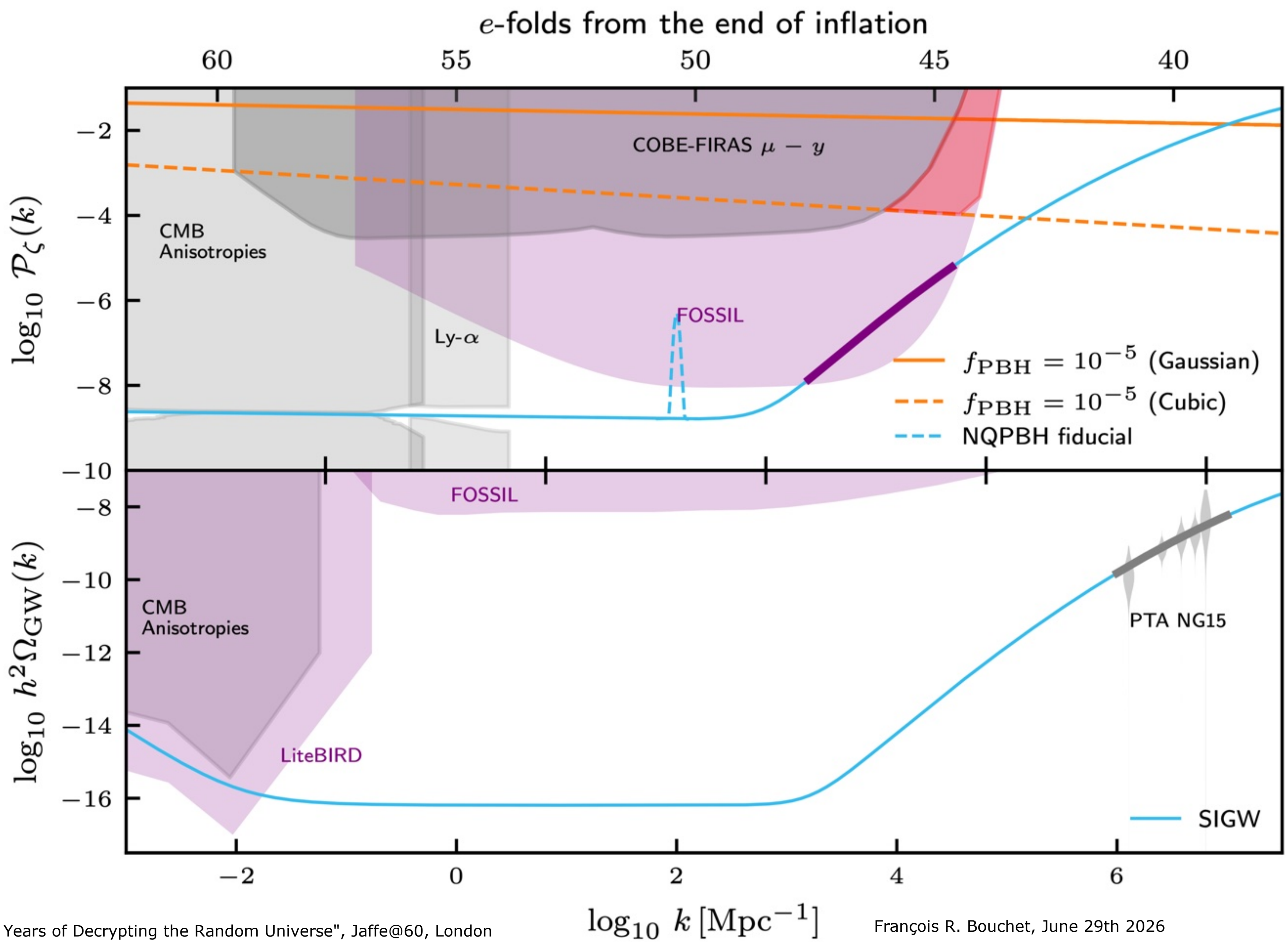
- ~130 frequency bands covering 50 GHz - 2 THz → Ideal for the challenge of foregrounds
- Monopole **y distortion >3 orders magnitude better** than FIRAS → **$\sigma(y)=5.\ 10^{-9}$**
- Average temperature of hot gas from relativistic SZ down to **$kT_{eSZ} \simeq$ 1.3 keV at tens σ**
- μ distortions **hundreds of times better** than FIRAS → **$\sigma(\mu)=1.5.\ 10^{-8}$**

FOSSIL sensitivity versus signals



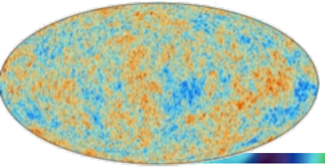
- Great μ and y science ahead
- Learning on Inflation, dark matter and BH formation to Galaxy formation, recombination, CIB and dust SED…
- If ESA selects it as M8 mission

FOSSIL GRASP on Primordial curvature and GW

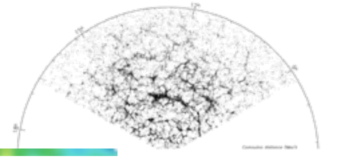


The forecasted sensitivity of FOSSIL is shown in pink.

Conclusions

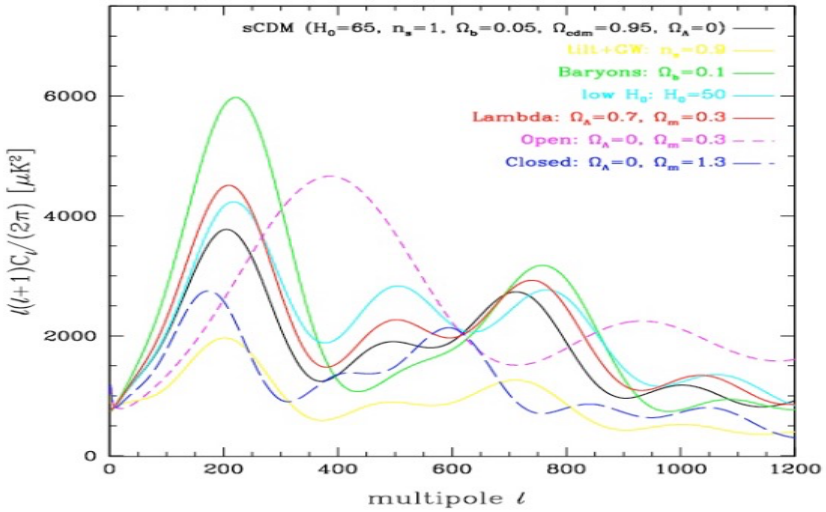


The magic of CMB is still alive...



- *CMB* can be accurately measured, AND:
  - compared to precise theoretical predictions
  - with a rich phenomenology
  - in a statistically reliable, and
  - computationally tractable way

("Bayesian Methods in Cosmology », 2010)



- There are very few situations in cosmology, astrophysics (or indeed physics) where all of these conditions are met.

- It is the intersection of these qualities that makes CMB such a powerful cosmological probe!

→ CMB will remain an extraordinary fruitful endeavour for decades to come

Wishing many more years of fun with our random Universe to Andrew